

T²EA: Target-Aware Taylor Expansion Approximation Network for Infrared and Visible Image Fusion

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Abstract—In the image fusion mission, the crucial task is to generate high-quality images for highlighting the key objects while enhancing the scenes to be understood. To complete this task and provide a powerful interpretability as well as a strong generalization ability in producing enjoyable fusion results which are comfortable for vision tasks (such as objects detection and their segmentation), we present a novel interpretable decomposition scheme and develop a target-aware Taylor expansion approximation (T²EA) network for infrared and visible image fusion, where our T²EA includes the following key procedures: Firstly, visible and infrared images are both decomposed into feature maps through a designed Taylor expansion approximation (TEA) network. Then, the Taylor feature maps are hierarchically fused by a dual-branch feature fusion (DBFF) network. Next, the fused map of each layer is contributed to synthesize an enjoyable fusion result by the inverse Taylor expansion. Finally, a segmentation network is jointed to refine the fusion network parameters which can promote the pleasing fusion results to be more suitable for segmenting the objects. To validate the effectiveness of our reported T²EA network, we first discuss the selection of Taylor expansion layers and fusion strategies. Then, both quantitatively and qualitatively experimental results generated by the selected SOTA approaches on three datasets (*MSRS*, *TNO*, and *LLVIP*) are compared in testing, generalization, and target detection and segmentation, demonstrating that our T²EA can produce more competitive fusion results for vision tasks and is more powerful for image adaption. The code will be available at <https://github.com/MysterYxby/T2EA>.

Index Terms—Image fusion, Taylor expansion approximation, infrared and visible feature representation, dual-branch feature fusion, object detection.

I. INTRODUCTION

IMAGE fusion has attracted massive research attention in the field of computer vision, aiming at combining multi-model images in the same scene to generate an image that can describe the scene accurately and comprehensively [1], [2]. Among various source image types, the fusion of infrared and visible image matches with the practical application of most public security [3]. The infrared sensor reflects the target information by capturing the thermal radiation in the scene, and can identify infrared targets such as people and vehicles even under adverse conditions like low illumination, occlusion and hiding [4]. However, infrared images usually suffer from low resolution and poor scene texture details. By contrast, visible sensor gathers the reflected light of objects, and visible images are suitable for human visual perception because of their high spatial resolution and considerable texture information, but they are easily affected by rain, fog and other objective environmental factors, so the image quality is unstable [5]. Due to the limitations of the physical properties, imaging principles, and observation perspectives of the sensors, it is difficult for a single image sensor to extract sufficient effective information from the scene that the generated images often exist the limitations of scene expression. Such images are not beneficial for vision tasks in complex scenarios, such as target detection [6], [7], [8], object segmentation [9] and classification [10], and tracking [11], [12], [13]. Therefore, image fusion is particularly important in utilizing multiple image information from sensors at the same scene to output a fused image that is more suitable for human visual perception or further processing and analysis. Image fusion technology can significantly overcome the shortcomings of a single sensor and enhance the visibility of outputs which are beneficial for us to comprehensively and accurately obtain the information of targets and scenes.

Traditional approaches first decompose images into multi-scale feature maps [14], [15], [16], [17], [18], then design fusion rules to fuse them, and finally reconstruct the fused images using the inverse decomposition methods. These methods have strong mathematical reasoning, powerful interpretability, and can achieve pleasing results in specific

Received 4 June 2024; revised 5 November 2024 and 12 December 2024; accepted 28 December 2024. Date of publication 1 January 2025; date of current version 7 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42371321 and in part by the Natural Science Foundation of Wuhan Donghu University under Grant 2024dhzk004. This article was recommended by Associate Editor M. Zhou. (Corresponding authors: Qian Li; Yansheng Li.)

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Digital Object Identifier 10.1109/TCSVT.2024.3524794

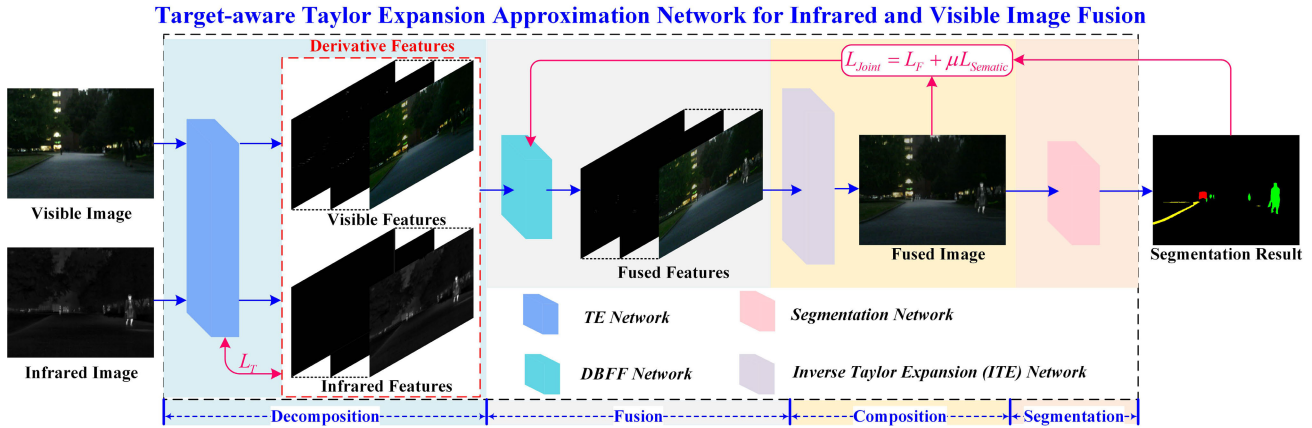


Fig. 1. The flowchart of the proposed T²EA network.

scenarios. But they are not scene adaptation due to their single strategy for extracting and fusing image features, resulting in the unfavorable outputs with low illumination, distortion, and artifacts. The deep learning based methods [19], [20], [21], [22] convert the image pixel values into their weight relationships and use loss functions to determine their weight attention with their powerful feature extraction and data representation learning ability. These end-to-end schemes overcome the shortcomings (such as insufficient feature extraction and inability in different scenarios adaption) of traditional methods. However, due to their black-box nature, their success lacks interpretability and the final results rely too heavily on the loss function. More importantly, these methods ignore the correlation among the pixels, resulting in the loss of important global textures in the fusion results, which cannot simultaneously balance target highlighting and scene enhancement, and are not efficient for the visual task of object detection. In response to the above-mentioned challenges, we propose a target-aware Taylor expansion approximation (T²EA) network which is a novel perspective for infrared and visible image fusion. Specifically, the reported T²EA includes two tasked parts: image fusion in the framework of Taylor expansion approximation and object segmentation, of which the flowchart is shown in Fig. 1. From the figure, one can see that there are three tasks, including decomposition for image separation, feature fusion at the same level, and multi-level feature composition with an inverse Taylor expansion (ITE) network, in image fusion. The segmentation network is explored to further optimize the fused results which are beneficial for interesting object segmentation.

Our contributions are listed as follows:

- 1) A target-aware Taylor expansion approximation network is proposed for infrared and visible image fusion, where a new interpretable decomposition perspective is developed for image fusion based on Taylor expansion.
- 2) A dual-branch feature fusion network with attention mechanisms is reported to fuse the decomposed features with a Taylor expansion (TE) network optimized by a novel loss function.
- 3) A jointing loss function (fusion and semantic segmentation) is developed to refine the fusion features for object segmentation.

The remainder sections are structured as follows: Section II reviews the related works of image fusion in detail. Section III introduces each component (decomposition, fusion, composition, and segmentation) of our target-aware Taylor expansion approximation (T²EA) network. We provide the experimental results and their analysis to validate the effectiveness of our method in Sect. IV and summarize our work in Sect. V.

II. RELATED WORK

A. Traditional Fusion Methods

In traditional fusion methods, multi-scale decomposition strategy is the most common employed method to separate an original image into multi-scale sub-images. These multi-scale decomposition based approaches have three procedures: (1) separating an original image by a multi-scale decomposition (MSD) method, (2) fusing sub-images by a handcrafted fusion rules, and (3) obtaining the final fusion results by the inverse transform of the MSD. That is to say, the multi-scale decomposition algorithms (such as Guided image filtering (GIF [23]), Multi-scale decomposition (MSD [24]), Pyramid decomposition (PD [25]), Discrete wavelet transform (DWT [26]), and Non-subsampled contour transform (NSCT [27])) and fusion rules [28] are the two main factors affecting the fusion results. For instance, Li et al. [29] exhibited a jointly clustered patch online dictionary learning (JCPORDL) method based on guided filter and online robust dictionary learning to construct a lightweight but practical dictionary and process large-scale and dynamic data. Wang et al. [30] presented a fusion method of infrared and visible image sequences via a joint low-rank and sparse decomposition (JLRSD) model. Xu et al. [31] introduced a multi-stage fusion algorithm (MSPIF) for infrared and visible images based on DWT and Retinex theory, to achieve high-quality visualization results. Li et al. [32] proposed an infrared and visible image fusion scheme in nonsubsampling contourlet transform (NSCT) domain. Although these approaches yielded promising fusion results, they still have drawbacks, such as: (1) image features extracted by a single decomposition strategy are insufficient for preserving the global details. (2) Handcrafted fusion schemes are task-specialized designed, which are not adaptive for different scenarios.

B. Deep Learning Based Approaches

In recent years, deep learning has been widely applied in computer vision due to its powerful capability in feature extraction and data representation. To address the shortcomings of traditional methods, deep learning based fusion approaches replace handcrafted designs by training fusion models to learn feature representation. These methods can be roughly categorized into four types: Convolutional neural networks (CNNs [33], [34], [35], [36], [37]), Auto-encoder networks (AENs [38], [39], [40], [41]), Generative adversarial networks (GANs [42], [43], [44], [45], [46]), and Transformer-based methods (TMs [47], [48], [49], [50]).

1) *CNNs*: As one of the most common models in deep learning, they map images from high-dimensional data to deep image features in non-linearity. They train networks to obtain optimal net parameters and generate fused images by designing different network architectures and loss functions, there is no need to design complex fusion strategies. Such as: Xu et al. introduced a unified framework (U2Fusion [51]) to adaptively preserve important information in infrared and visible images. Ma et al. [52] designed a fusion framework based on salient object detection to generate fused images in real scenes. Wang et al. [53] proposed an unsupervised cross-modal “registration-fusion” joint framework (UMF-CMGR) to enhance the generalization performance of fused images in complex scenes. To synthesize a single high-visualization fused image for facilitating high-level vision tasks, Tang et al. [54] developed a semantic-aware real-time image fusion network (SeAFusion). Li et al. [55] reported a representation learning guided fusion framework, termed as LRRNet, by formulating a fusion task mathematically and establishing a connection between its optimal solution and the network architecture to specify a good fusion architecture. Liu et al. [56] proposed a multi-interactive feature learning architecture for image fusion and segmentation, namely SegMiF, and exploited dual-task correlation to promote the performance of both tasks. Dong et al. [57] proposed a multi-level feature injection infrared and visible image fusion method based on an image decomposition model, termed as MFIFusion, to fully utilize the information of features in network layers. These end-to-end methods overcome the insufficiency of feature extraction of handcrafted fusion approaches, but their fusion results excessively depend on the design of loss functions, resulting in the deficiency of the model’s constraints and weak generalization ability.

2) *AENs*: To improve the efficiency of the above CNN-based fusion methods, AEN-based approaches first train encoders on large datasets to extract image features. Then, the extracted features are fused using handcrafted fusion strategies. Finally, a trained decoder is employed to reconstruct the fused features for the fusion result. For example, Li et al. [58] developed a fusion algorithm combined with dense structures (DenseFuse) to improve fusion performance. Zhang et al. [59] proposed a universal image fusion framework to enhance generalization ability. Xu et al. [60] proposed a decomposition method for visible and infrared image fusion (DRF) to alleviate the problem of inappropriate extraction of unique information. Zhao et al. [61] proposed a Correlation-Driven feature Decomposition Fusion (CDDFuse)

network to tackle the challenge in modeling cross-modality features and decomposing desirable modality-specific and modality-shared features. Xu and Xin [62] developed a dual-branch feature decomposition fusion (DAF) network with domain adaptive to preserve key features during the fusion process. However, the feature fusion schemes of them are manually designed, which limits their adaption of different scenarios.

3) *GANs*: GAN-based methods use the competition between the discriminator and the generator to effectively reduce the dependence on “Ground-truth” images. For example, Ma et al. [63] established an adversarial game between the generator and the discriminator to generate fused images with complementary information between infrared and visible images (FusionGAN). Li et al. [64] improved the structure of the generator and discriminator by introducing attention mechanisms to generate images with richer details. Ma et al. [65] further designed a GANs framework with multi-class constraints (GANMcC) to maintain clear contrast and sufficient texture. These GAN-based methods can transfer as many rich details as possible from the original images to fusion results by learning local features to improve the generalization ability of fusion networks, but they neglect the extraction of long-range correlations in images, resulting in the loss of important global textures in the fusion results.

4) *TMs*: In order to extract long-range correlations in original images, TM-based approaches introduce a multi-head self-attention mechanism into AENs to construct an advanced model architecture. For instance, Ma et al. [66] proposed a novel image fusion framework based on transformer to preserve texture, structure, and intensity information. Li and Wu [67] reported a cross attention mechanism (CAM) to enhance the complementary information and developed a cross attention mechanism based infrared and visible image fusion approach (CrossFuse). Huang et al. [68] proposed a progressive information interaction fusion network based on Transformer to remove harmful information from source images. Song et al. [69] proposed an image fusion model based on Transformer and an improved DenseNet network module to reduce edge blurring and information redundancy. They produce high-quality fusion images with enjoyable structures at the expense of a heavy time cost.

III. TARGET-AWARE TAYLOR EXPANSION APPROXIMATION (T²EA) NETWORK

In this section, each component, including the Taylor expansion approximation network, dual-branch feature fusion network and target-aware fusion feature refinement training strategy, of our target-aware Taylor expansion approximation (T²EA) network is presented.

A. Taylor Expansion Approximation Network

1) *Taylor Expansion Theory*: Taylor expansion, as an important mathematical tool [70], has been widely applied in computer vision tasks, such as image restoration [71], optical flow estimation [72], [73] and denoising model construction [74]. According to Taylor expansion theorem, when $f(x)$ ($x \in (a, b)$) has an n -order derivative, then it can

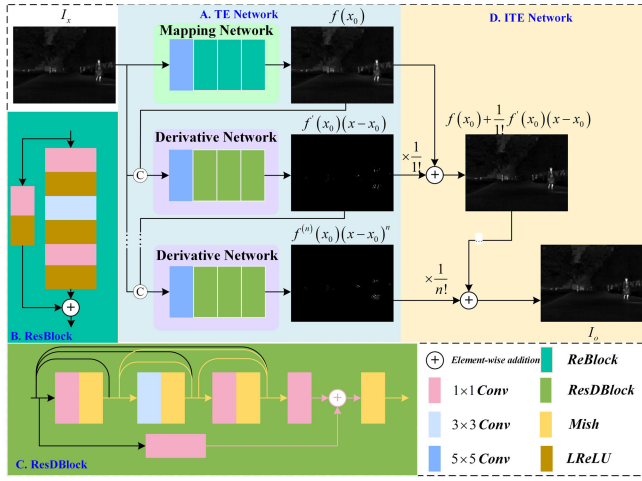


Fig. 2. The architecture of Taylor expansion approximation network.

be unfolded at x_0 ($x_0 \in (a, b)$) as follows:

$$\begin{aligned} f(x) &= f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \\ &= f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k, \end{aligned} \quad (1)$$

where $!$ denotes the factorial operation. $f(x)$, $f(x_0)$ and $\sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ represent the source **visible or infrared** image, the basic large-scale feature map, and the derivative features, respectively. From the Taylor expansion formula (1), one can see that its feature maps among all levels have a close correlation, which will be helpful for networks to transfer global multi-structures from original images to fused results. To fully extract image feature maps, we develop a mapping network to approximate the basic layer while a sharing derivative network for the derivative features.

2) Network Architecture for Taylor Expansion Approximation: In Eq. (1), there are two main components including the base layer $f(x_0)$ and its derivative layers (such as $f^k(x_0)$, $k \in [1, n]$) to be estimated. To sufficiently extract their features, we explore a mapping network $\Phi_b(\cdot)$ to calculate the base layer (which can be represented as $f(x_0) = \Phi_b(f(x))$) and a derivative network $\Phi_d(\cdot)$ for each derivative component. As we can see from Eq. (1) that its neighboring derivative components have a close correlation. In addition to the latter being the derivative of the former, there is also an additional term $(x - x_0)$ which is related to $f(x)$ and $f(x_0)$. To address this issue, we concatenate $f(x)$ and $f^k(x_0)$ ($k \in [1, n]$) into Φ_d as an input for inference, which can be expressed as

$$f(x) = \Phi_b(f(x)) + \sum_{k=1}^n \frac{\Phi_d(\text{Concat}(f^{(k-1)}(x_0), f(x)))}{k!}, \quad (2)$$

whose architecture is shown in Fig. 2.

a) Mapping network: The network shown in Fig. 2(A) extracts global information from images to prevent the loss of large-scale details, from which we can see that it is composed of a convolutional layer with kernel size of 5×5 and three ResBlocks. The specific framework of ResBlock is shown

in Fig. 2(B), including three convolutional layers with kernel size of 1×1 , a convolutional layer with kernel size of 3×3 , and four Leaky ReLU (LReLU) layers after each convolutional layer to enhance network stability.

b) Derivative network: The network consists of two convolutional layers with 5×5 kernel size and three ResDBlocks. Figure 1(C) illustrates the ResDBlock's specific framework. To retain as many fine features as possible during feature extraction, dense connections and the *Mish* activation function are incorporated into the ResBlock.

3) Loss Function: In order to obtain the desired Taylor decomposition feature maps, we employ a Taylor loss

$$L_T = L_{\text{pixel}} + \lambda L_{\text{grad}} \quad (3)$$

to optimize the Taylor expansion approximation network for optimal parameters of both the mapping and derivative networks, where L_{pixel} and L_{grad} represent the pixel loss and gradient detail loss, respectively. λ is a hyperparameter to constrain their magnitude difference.

The pixel loss L_{pixel} measures the difference between two images at the pixel level, which is defined as

$$L_{\text{pixel}} = \frac{1}{HW} \|I_x - I_o\|_1, \quad (4)$$

where I_x and I_o respectively represent the input image (infrared or visible images) and the output of the Taylor expansion approximation network. H is the height of the image, W is its width, and $\|\cdot\|_1$ represents the L_1 norm.

Generally, we expect that the feature maps decomposed by the network from the source image can be fully employed for the reconstructed result via the inverse Taylor transformation. Except for the pixel level loss, we propose a gradient loss

$$L_{\text{grad}} = \frac{1}{HW} \| |\nabla I_o| - \max(|\nabla I_x|, |\nabla I_n|) \|_1, \quad (5)$$

which is jointed to train the Taylor expansion approximation (TEA) network for net parameters which can represent as more information in the original image as possible, where ∇ , $|\cdot|$, and $\max(\cdot)$ denote the operations of Sobel, absolute value, and maximum value, respectively. I_n represents the feature maps decomposed by the derivative network (n is the number of feature maps (Taylor layers), which is a hyperparameter and will be discussed in the ablation studies). The TEA network is used to separate the original image into multi-layer feature maps by optimizing the jointed loss in Eq. (3).

B. Dual-Branch Feature Fusion (DBFF) Network

1) Architecture: To generate high-quality fused images from infrared and visible images, we introduce a dual-branch feature fusion (DBFF) network shown in Fig. 3, from which one can see that there are two parallel branches (which have the same architecture) to extract features from infrared and visible feature maps, respectively. Each branch includes a convolutional layer and two gradient residual dense block (GRDB) modules for effectively extracting shallow features. GRDB, as shown in Fig. 3, derives from ResBlock but is different from it, including two branches (lightweight dense and gradient residual networks). In the lightweight dense residual network, there are two 3×3 convolutional layers and one 1×1 convolutional layer to fully extract features with dense connections.

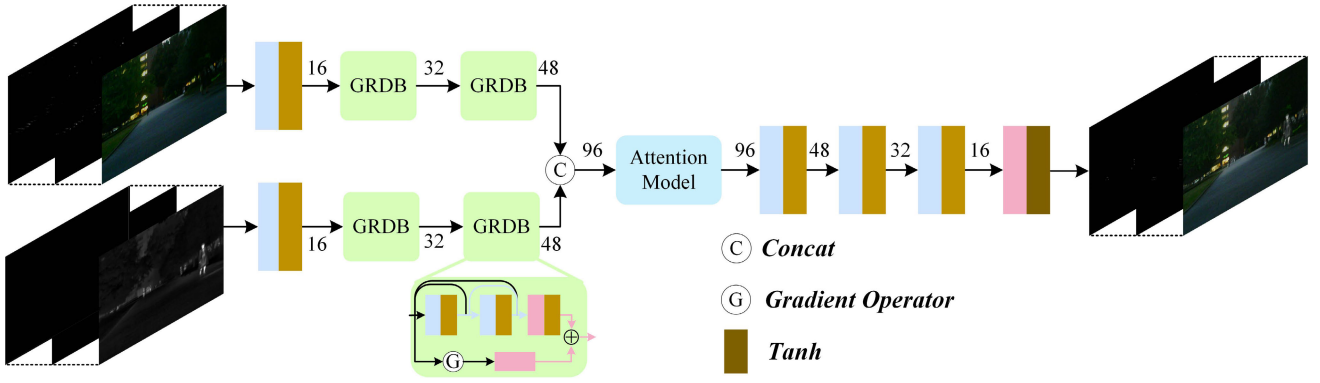


Fig. 3. The framework of the dual-branch feature fusion (DBFF) network.

The gradient residual network calculates gradient maps to assess feature magnitudes and uses a 1×1 convolutional layer to standardize channel dimensions. Finally, the two extracted shallow features are added via an element-wise addition for fusion. Note that the LReLU activation function is used after each convolutional layer.

After concatenating infrared and visible features at the same layer of Taylor expansion, we employ an attention model, three consecutive 3×3 convolutional layers (the activation function after each layer is LReLU), and a 1×1 convolutional layer (its following activation function is Tanh). As the above-mentioned discussion, the self-attention mechanism in transformer effectively addresses long-range dependencies, by considering its complexity, we adopt the spatial-channel attention mechanism (CBAM) which is detailedly introduced in the literature [75]. Padding in the fusion network is set identically, and the stride is set to 1. Since our network does not involve in any downsampling, the size of the fused image is consistent with that of the source images.

2) *Loss Function*: To fully leverage the effectiveness of fusion model and generate high-quality fused images, the loss function

$$L_F = L_{int} + \alpha L_{tex} \quad (6)$$

is utilized to optimize network parameters, where L_{int} denotes an intensity loss, which constrains the overall intensity of the fused image and can be formulated as

$$L_{int} = \frac{1}{HW} \|I_f - \max(I_r, I_{vis})\|_1, \quad (7)$$

where I_f , I_r and I_{vis} denote the fused, infrared and visible images, respectively. L_{tex} represents a texture detail loss and is defined as

$$L_{tex} = \frac{1}{HW} \|\nabla I_f - \max(|\nabla I_r|, |\nabla I_{vis}|)\|_1 \quad (8)$$

to learn network parameters which can well represent fine detail information and transfer them from the original images to the fused image. And α is a weighting parameter to balance their difference. With the employment of the jointed loss function, the DBFF network can simultaneously achieve the optimal intensity and texture distributions for the pursuit of the details of both the targets and the scene.

C. Target-Aware Fusion Feature Refinement

The aim of image fusion is not only for enhancing the understanding of the scene, but the objects are also highlighted in the fused results which are beneficial for detection methods to accurately detect them. In view of this point, we further exploit a segmentation network (in our work, we employ the improved BisNet (IBisNet) segmentation network involved in the Ref. [76]) to refine fusion features. As our task is to optimize the DBFF's parameters, the segmentation network parameters are fixed to be the same as those in the IBisNet when training the DBFF network.

1) *Loss Function*: To train the DBFF network for an enjoyable visuality with rich textures and highlighted objects, a joint loss, including the fusion loss L_F and a proposed semantic segmentation loss $L_{semantic}$,

$$L_{Joint} = L_F + \mu L_{semantic}, \quad (9)$$

is designed for pursuing optimal parameters. In Eq. (9), μ is a changeable hyperparameter to balance the fusion loss and the semantic loss, which will be discussed in Sub-sect. III-C2 in detail, and $L_{semantic}$ is formulated as

$$L_{semantic} = L_{main} + \beta L_{aux}, \quad (10)$$

where β is utilized to eliminate the magnitude differences, and two sub-semantic losses, including the main semantic loss L_{main} and the auxiliary semantic loss L_{aux} , are respectively defined as:

$$\begin{cases} L_{main} = -\frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W \sum_{c=1}^C I_{so}^{(h,w,c)} \log(I_{sm}^{(h,w,c)}) \\ L_{aux} = -\frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W \sum_{c=1}^C I_{so}^{(h,w,c)} \log(I_{sa}^{(h,w,c)}), \end{cases} \quad (11)$$

where $I_{so} \in \mathbb{R}^{C \times H \times W}$ represents the one-hot vector derived from the segmentation label $I_{label} \in (1, C)^{H \times W}$. The segmentation network generates the main segmentation result $I_{sm} \in \mathbb{R}^{C \times H \times W}$ and the auxiliary segmentation result $I_{sa} \in \mathbb{R}^{C \times H \times W}$ to optimize the parameters of the DBFF network.

2) *Multi-Stage Training Strategy*: Due to the limitation of the computational capacity in our laboratory's computer device, a multi-stage training strategy is exploited to obtain a pleasing fusion performance, including:

Stage 1: TEA Network: To obtain optimal multi-layer Taylor feature maps, the original images are decomposed by training the Taylor expansion approximation (TEA) network (whose architecture is shown in Fig. 2) with the employment of the loss function (3).

Stage 2: TEA + DBFF: Its architecture includes Decomposition (TE network in Fig. 2(A)) + Fusion (DBFF network in Fig. 3) + Composition (ITE network in Fig. 2(D)) in Fig. 1. In this stage, the parameters of the TEA network are fixed, we train the DBFF network to obtain net parameters for fusion results by the loss function (6).

Stage 3: TEA + DBFF + Segmentation: To further refine the fusion parameters of the DBFF network for target segmentation, we employ a high-level visual task (IBisNet segmentation [76]) and incorporate it into the scheme of Stage 2. In the training stage, the parameters of both TEA and the segmentation network are fixed for the pursuit of the DBFF network parameters. The loss function (9) is utilized to train the DBFF network. It is worth noting that the hyperparameter μ in the loss function are dynamically adjusted with the number of iteration, which can be formulated as

$$\mu = INT(iter/10) + 1, \quad (12)$$

where $INT(\cdot)$ and $iter$ represent the numerical rounding operation and the number of iteration, respectively. The fusion network is expected to generate fusion images that adapt to the segmentation task, with the increment of iteration, the fusion model becomes better, the semantic loss (which becomes more important, illustrating that μ becomes larger) can guide the fusion network to be trained more suitable for target segmentation.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To verify the effectiveness of our T²EA network, we arrange the experimental parts as follows: experimental preparation for testing, ablation studies to discuss the selection of Taylor expansion layer and fusion strategies, comparative experiments to validate their SOTA performance and generalization ability, vision tasks on their fusion results to demonstrate further application, and computational complexity to verify their efficiency.

A. Experimental Preparation

1) Environment and Dataset: We verify the feasibility of our T²EA network on the MSRS [77] dataset, in which 1038 infrared and visible image pairs (a total of 2076 images with 640×480 pixels) are selected for training. All the experiments were implemented using Pytorch software on a PC with Intel(R) Xeon(R) CPU E5-2620 V3 @ 2.40GHZ 16GB memory associated with a NVIDIA Tesla P100 GPU.

2) Parameters Setting: We train our T²EA network in three stages presented in Sub-sect. III-C2. In **Stage 1**, 2076 images (1038 infrared and visible image pairs) were utilized to train the TEA network. In both **Stage 2** and **Stage 3**, these images were also employed for training the DBFF network. The Adam optimization algorithm was utilized to optimize network performance with a learning rate being set to 0.00001. The parameter λ in Eq. (3) was empirically set to 0.3 and the parameter n in Eq. (5) would be discussed in ablation experiments. Referring to the literature [76], the parameters α in Eq. (6) and β in Eq. (10) were set to 10 and 0.1, respectively.

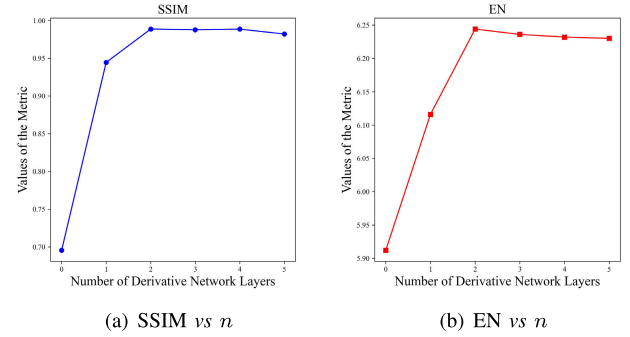


Fig. 4. The quantitative SSIM and EN values along with the number of derivative layers. **Note that SSIM is used to evaluate the TEA network while EN is to assess our T²EA under different Taylor expansion layers.**

3) The Selection of Compared Methods and Evaluation Indexes: To demonstrate the superiority of our T²EA, we compare it with seven SOTA fusion methods, including DenseFuse [58], DRF [60], FusionGAN [63], MFIFusion [57], SeAFusion [54], UMF-CMGR [53], CDDFuse [61], LRR-Net [55], and CrossFuse [67], in quantitation and qualification. Note that these selected SOTA fusion methods are trained on the same dataset and training environment. In objective evaluations, quantitative metrics, including entropy (EN [78]), standard deviation (SD [79]), mutual information (MI [80]), average gradient (AG [81]), visual quality fidelity (VIF [82]), and structural similarity (SSIM [83]), are employed to assess their performance. Specifically, the larger the EN and AG values, the richer the details. The higher the MI and SSIM values, the stronger the correlation between the fused images and the source images. And the higher the SD and VIF values, the higher the contrast for visualization.

B. Ablation Studies

1) The Impact of n on Reconstruction Results: To achieve optimal Taylor expansion feature maps, we use the SSIM value to analyze the influence of different parameters n on the output of the TEA network on the MSRS dataset by the “**Stage 1**” strategy introduced in Sub-sect. III-C2 and the EN value to discuss its impact on the final fusion result. Figures 4(a) and 4(b) respectively show the SSIM and EN values along with the change of parameter n , from which we can observe that the TEA network achieves the best SSIM while our reported T²EA method yields the best EN when $n = 2$, illustrating that the details of the output feature maps are the richest when the Taylor expansion layer is 2. Meanwhile, we also provide a visualization example in **Supplementary Materials** which can also validate such conclusion. Therefore, we set it to 2 in the following experiments.

2) The Discussion of Different Fusion Strategies: There are three fusion strategies in our work, including DBFF + Segmentation (DBFFSeg, which is first trained by the loss in Eq. (6) with the source images, and is then trained by the loss in Eq. (9) to fine tune the DBFF’s parameters), TEA + DBFF (TEADBFF, which is trained by the “**Stage 2**” strategy in Subsubsect. III-C2), and TEA + DBFF + Segmentation (T²EA, which is trained by the “**Stage 3**” strategy in Subsubsect. III-C2). To verify their superiority, we test them on the MSRS dataset (in which 361 pairs of visible

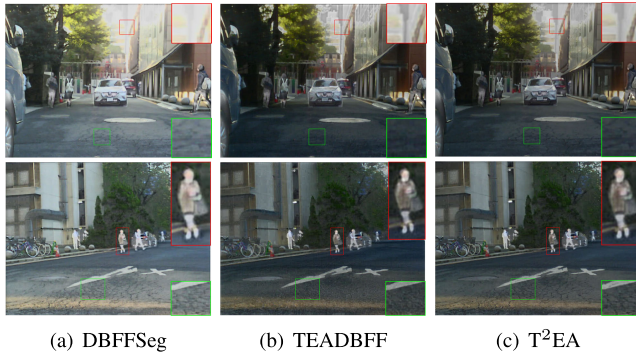


Fig. 5. The visualization results with different fusion strategies, (a) DBFF + Segmentation (DBFFSeg), (b) TEA + DBFF (TEADBFF) (Stage 2), and (c) TEA + DBFF + Segmentation (T²EA) (Stage 3, which is selected in our paper).

TABLE I
AVERAGE QUANTITATIVE METRICS GENERATED BY DIFFERENT FUSION STRATEGIES ON THE MSRS DATASET (THE BEST VALUE IS MARKED WITH RED FONT)

Methods	EN	MI	AG	SD	mIoU
DBFFSeg	6.159	1.642	2.234	25.333	69.15
TEADBFF	6.204	2.004	2.133	29.381	64.23
T ² EA	6.244	3.302	2.776	32.514	72.28

and infrared images are selected). Figure 5 shows two typical visual comparisons, from which we can find that all of them perform well on highlighting targets (e.g. human) in the fused results. However, the DBFFSeg and TEADBFF fusion schemes have their own drawbacks. For example, DBFFSeg uses single feature sets to produce fused results which are over-exposure, resulting in the loss of rich details in over-exposed areas (as shown in Fig. 5(a)). TEADBFF addresses the over-exposure problem by the usage a TEA network, but it generates low-contrast fusion results and fails to preserve details in dark areas (as shown in Fig. 5(b)). In contrast, T²EA can adaptively adjust the illumination of the fused results by the usage of both the TEA and sementation networks, which can suppress over-exposed areas and improve the contrast of the low-light areas when the intensity of the visible images are uneven (as shown in Fig. 5(c)). We also provide their quantitative evaluations in Table I. From the Table, one can see that the T²EA approach achieves the best quantitative performance, which demonstrates its superiority and is consistent with the analysis of visual comparisons. Moreover, we also provide the mIoU values produced by the segmentation method [76] on the fusion results, where one can see that our fusion result is more beneficial for object segmentation due to its best mIoU value. Meanwhile, we also provide lots of experimental results which are generated by these three fusion strategies and are further used for vision tasks in **Supplementary Materials**. From the comparisons, we can find that the T²EA network is superior to the two methods, so we select it as our reported method to be compared with other SOTA fusion methods in the following experiments.

C. Comparisons of the SOTA Fusion Methods

1) *Qualitative Evaluation*: Figures 6 and 7 provide the visual results generated by the selected SOTA fusion methods

TABLE II
QUANTITATIVE COMPARISONS OF SELECTED SCHEMES ON 361 PAIRS OF IMAGES IN THE MSRS DATASET

Methods	EN	MI	AG	SD	VIF	SSIM
DenseFuse [58]	5.937	2.643	2.058	23.568	0.692	0.911
DRF [60]	5.769	2.069	1.421	20.991	0.494	0.47
FusionGAN [63]	5.432	1.886	1.451	17.07	0.443	0.502
MFIFusion [57]	6.12	2.558	2.005	26.342	0.636	0.797
SeAFusion [54]	6.058	3.226	2.584	32.375	0.704	0.904
UMF-CMGR [53]	5.597	1.92	2.134	20.745	0.427	0.538
CDDFuse [61]	5.942	3.244	2.397	29.683	0.752	0.655
LRRNet [55]	5.945	3.944	2.405	29.725	0.755	0.905
CrossFuse [67]	5.9	2.42	1.918	27.521	0.757	0.833
Ours	6.244	3.302	2.776	32.514	0.778	0.913

on the “00016N” and “00024N” images from the MSRS dataset, from which we can notice that they can effectively yield fusion results, but they have their own merits and drawbacks in preserving details and highlighting targets as follows: (1) DRF (as shown in Figs. 6(d) and 7(d)) and FusionGAN (as shown in Figs. 6(e) and 7(e)) are capable of generating high-contrast fusion images and highlighting the targets, but many useful textures are missing, resulting in the blurry results. (2) Although DenseFuse (as shown in Figs. 6(c) and 7(c)) and MFIFusion (as shown in Figs. 6(f) and 7(f)) can transfer as many details as possible from visible images to fused results, they still over-smooth enjoyable structures in dark areas and the results are in low contrast. (3) By introducing a semantic network in the fusion framework, SeAFusion (as shown in Figs. 6(g) and 7(g)) produces enjoyable fusion results with salient objects. However, it still loses a lot of fine details of the visible images that leads to the low contrast of the fusion results for human visualization. (4) UMF-CMGR (as shown in Figs. 6(h) and 7(h)) performs well on highlighting targets (such as human) and improves the contrast of fusion results at the expense of details loss in low-brightness areas. (5) Both the objects in infrared images and enjoyable details in visible ones are highlighted in the fusion results produced by CDDFuse (as shown in Figs. 6(i) and 7(i)), but their contrast (which can be found in the enlarged red boxes) is still to be a failure to be improved. (6) Both LRRNet and CrossFuse pay more attention to preserving rich details in visible images, resulting in the loss of many fine details in infrared images, especially the under-exposure of targets (as shown in Figs. 6(j) and 7(j) as well as 6(k) and 7(k)). Compared to the above-discussed fusion approaches, our T²EA method generates the best fusion results which are more comfortable for human visualization due to that more details in visible images are preserved while retaining the highlighted targets in infrared images (as shown in Figs. 6(l) and 7(l)).

2) *Quantitative Assessment*: Their quantitative results are shown in Table II, from which we can observe that our T²EA approach the best value of all quantitative metrics, further demonstrating that our T²EA approach has a powerful ability in preserving details and highlighting objects of the fusion results for favorable visualization.

D. Generalization Experiments

To validate the generalization ability of the above-mentioned SOTA fusion methods, we respectively select 21 and 177 pairs

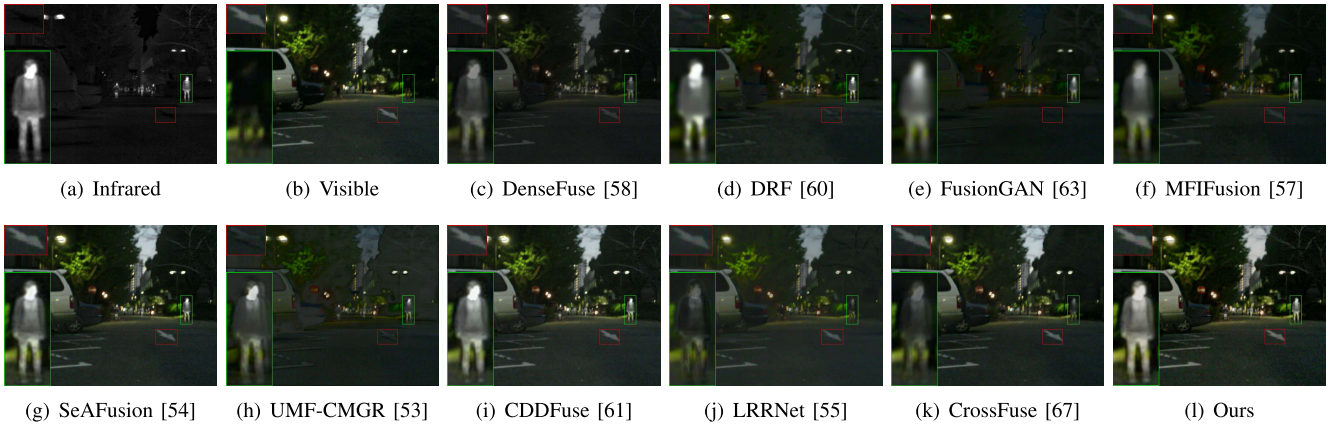


Fig. 6. Visual comparisons on the “00016N” image from the *MSRS* dataset.

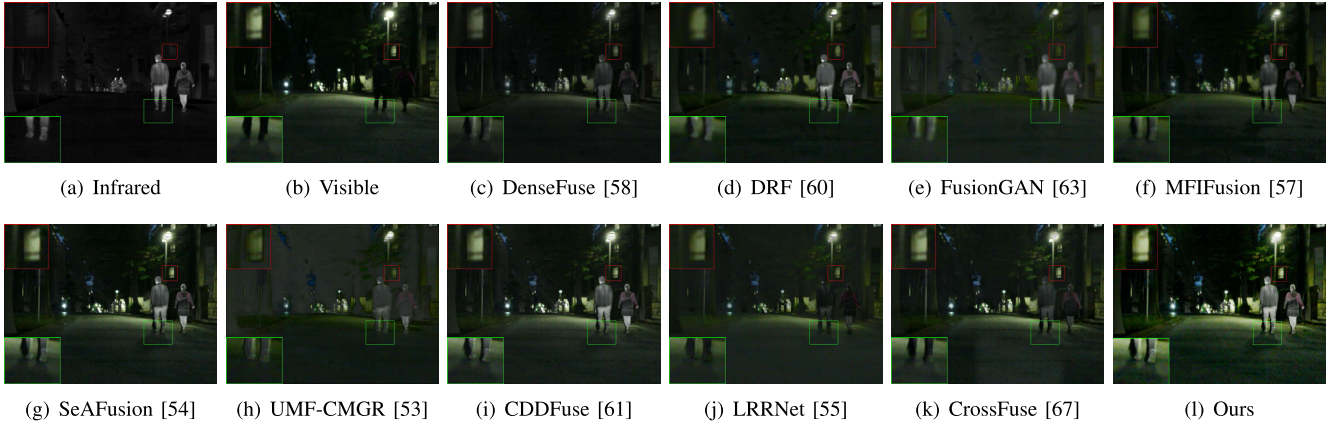


Fig. 7. Visual comparisons on the “00024N” image from the *MSRS* dataset.

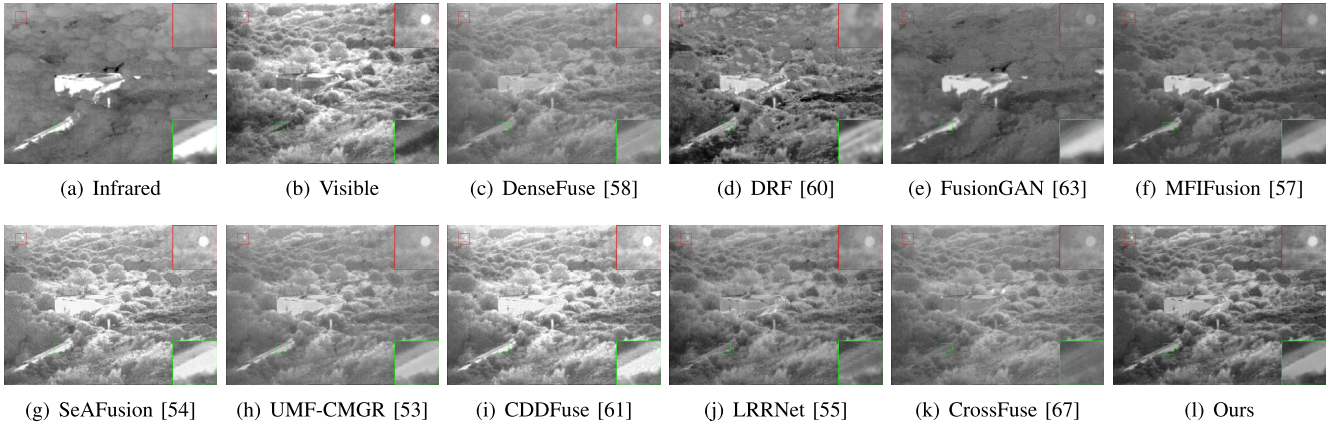
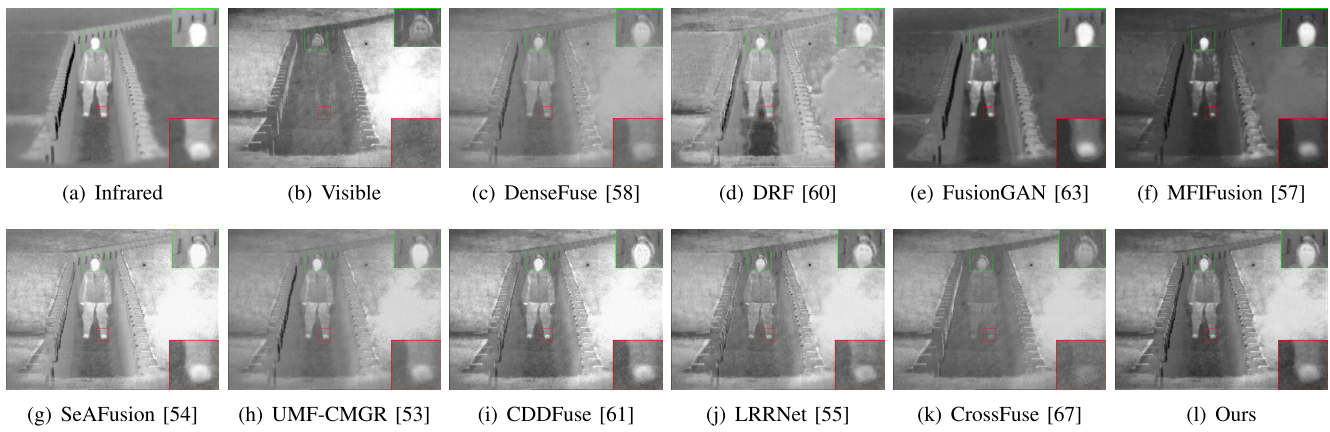
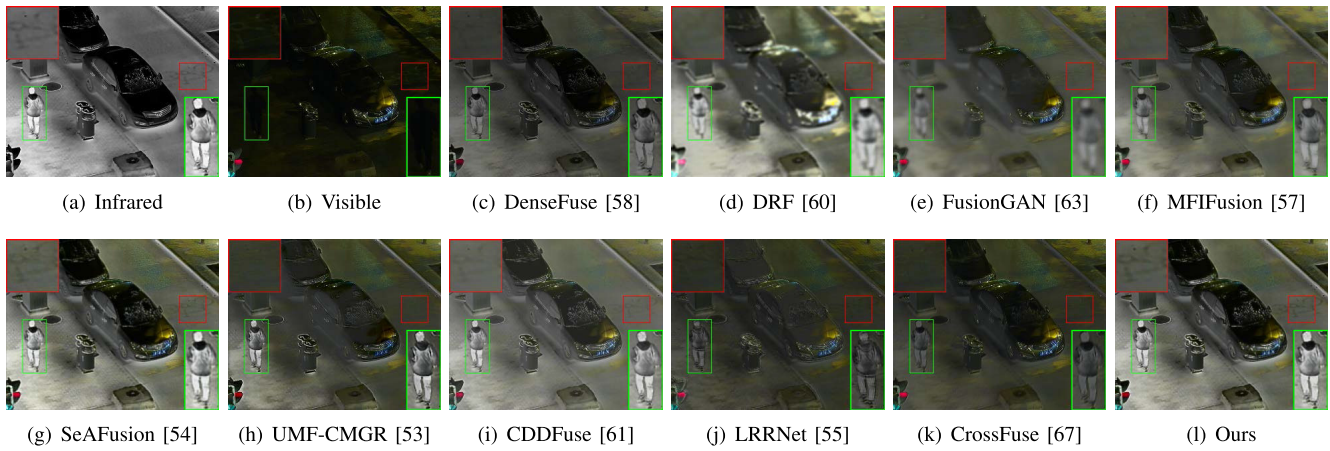
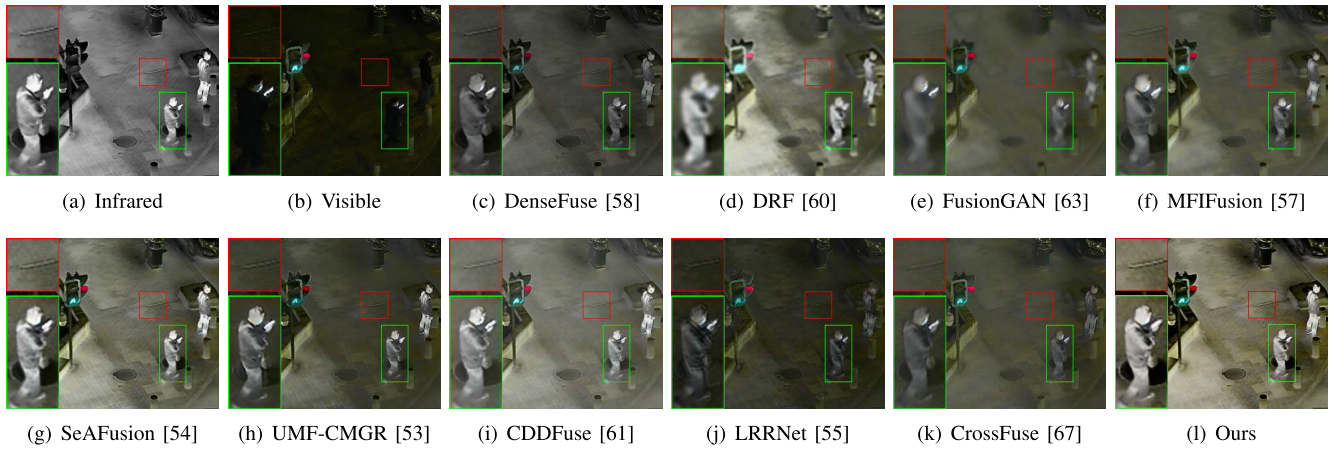


Fig. 8. Visual comparisons on the “Bunker” image from the *TNO* dataset.

of infrared and visible images from the *TNO Human Factors* [84] and *LLVIP* [85] datasets to test their image adaption and evaluate them in qualification and quantification.

1) *Subjective Evaluation*: Figures 8 and 9 present the visualization results by applying the SOTA fusion methods to the “Bunker” and “Meeting_016” images from the *TNO* dataset while Figures 10 and 11 are the ones on the “230024” and “200090” images from the *LLVIP* dataset. From the results, some observations can be concluded as follows: (1) DRF can produce high-contrast fusion results, but loses a lot of significant details of the visible images, such as the white

target in the red box of Fig. 8(d) and edges of the human in Figs. 9(d), 10(d), and 11(d). (2) Fusion results generated by FusionGAN and MFIFusion are over-smoothed. As shown in Figs. 8(e) to 11(e) and 8(f) to 11(f), both rich structures in dark areas and edges of highlighted targets are blurred. (3) As the same discussion in Sub-sect. IV-C, SeAFusion yields pleasing fusion results, in which the objects are well highlighted (which can be seen in Figs. 10(g) and 11(g)). However, due to the excessive emphasis on the large-scale objects, the fine small-scale structures may be smoothed in the fusion results, which can be found in the green box

Fig. 9. Visual comparisons on the “Meeting_016” image from the *TNO* dataset.Fig. 10. Visual comparisons on the “230024” image from the *LLVIP* dataset.Fig. 11. Visual comparisons on the “200090” image from the *LLVIP* dataset.

of Fig. 9(g) and the red boxes of Figs. 10(g) and 11(g). (4) DenseFuse and UMF-CMGR highlight targets in the fused results at the cost of contrast reduction on day-time image fusion (as shown in Figs. 8(c) to 9(c) and 8(h) to 9(h)) and over-smoothed details in low-brightness areas (as shown in Figs. 10(c) to 11(c) and 10(h) to 11(h)) on night-time image fusion. (5) CDDFuse tries its best effort to highlight the targets and preserve favorable details in fusion results shown in Figs. 8(i) to 11(i). However, it is not

good at preserving fine details in over-exposure areas shown in Figs. 9(i) and 9(j). (6) The results shown in Figs. 8(j) to 11(j) produced by LRRNet and Figs. 8(k) to 9(k) generated by CrossFuse are low contrast, of which the rich details in low-light areas are unable to be distinguished (as shown in the enlarged red box of Figs. 10(j), 11(j), and 10(k)). Contrarily, our T²EA scheme is image adaption to preserving details, improving contrast, and highlighting targets (as shown in Figs. 8(l) to 11(l)).

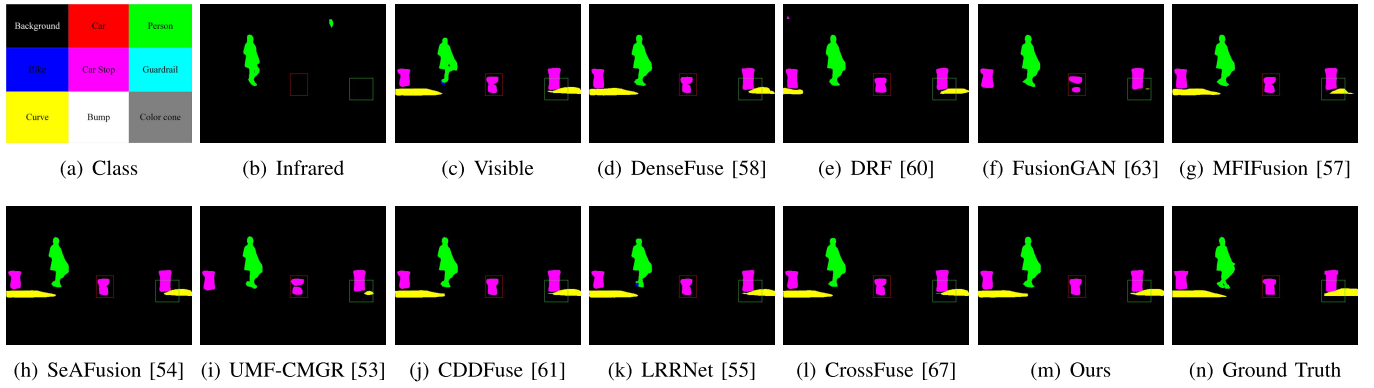


Fig. 12. Segmentation results with the IBisNet network on the fused images generated by applying the selected SOTA fusion methods to the “00123D” image from the *MSRS* dataset.

TABLE III
QUANTITATIVE COMPARISONS OF SELECTED SCHEMES
ON 21 PAIRS OF IMAGES IN THE *TNO* DATASET

Methods	EN	MI	AG	SD	VIF	SSIM
DenseFuse [58]	6.174	2.135	2.353	22.546	0.534	0.821
DRF [60]	6.714	1.742	2.424	30.621	0.362	0.521
FusionGAN [63]	6.363	2.249	2.205	26.068	0.413	0.664
MFIFusion [57]	6.542	2.233	2.412	30.286	0.507	0.839
SeAFusion [54]	6.862	2.776	4.718	38.036	0.629	0.914
UMF-CMGR [53]	6.329	2.138	2.773	26.561	0.556	0.828
CDDFuse [61]	6.839	2.782	4.444	34.295	0.615	0.51
LRRNet [55]	6.838	2.515	3.6	39.499	0.551	0.821
CrossFuse [67]	6.823	2.125	3.717	37.42	0.625	0.879
Ours	6.895	2.805	4.726	38.041	0.631	0.922

TABLE IV
QUANTITATIVE COMPARISONS OF SELECTED SCHEMES
ON 177 PAIRS OF IMAGES IN THE *LLVIP* DATASET

Methods	EN	MI	AG	SD	VIF	SSIM
DenseFuse [58]	6.821	3.337	4.111	32.878	0.601	0.864
DRF [60]	7.247	2.753	4.005	43.682	0.372	0.529
FusionGAN [63]	6.589	2.694	3.136	28.744	0.359	0.559
MFIFusion [57]	6.968	2.914	3.857	37.794	0.489	0.762
SeAFusion [54]	7.302	3.702	5.778	51.098	0.599	0.892
UMF-CMGR [53]	6.744	2.979	4.73	34.143	0.443	0.722
CDDFuse [61]	7.01	4.13	5.156	38.557	0.626	0.584
LRRNet [55]	7.01	4.33	5.156	38.557	0.666	0.877
CrossFuse [67]	6.495	2.627	4.021	27.086	0.555	0.736
Ours	7.374	4.133	5.514	51.941	0.632	0.895

2) *Objective Assessment*: Tables III and IV provide the average quantitative metrics produced by the SOTA fusion methods on the *TNO* and *LLVIP* datasets, respectively. From the tables, we can find that our T²EA method achieve the best values of both EN and SSIM metrics, further validating its superiority to others in preserving rich details from both infrared and visible images to the fusion results, which is consistent with the visual comparisons. In summary, quantitative evaluation associated with qualitative assessment validates the powerful ability of our T²EA approach in image adaptation and generalization, making it well-suited for extensive vision tasks.

E. Fusion Results Applied to Vision Tasks

In this sub-section, the fused results generated by the above-compared SOTA approaches are utilized

to be experimented for vision tasks (segmentation and detection), where the results can be employed to further verify their capacity in extensive applications.

1) *Segmentation Task*: We employ the improved BisNet (IBisNet) network [76], which is a pre-trained model, on the *MSRS* dataset (which includes a ground-truth classification dataset) to segment the targets from fusion results. The visual segmentation results produced by the IBisNet method on the infrared, visible, and fusion images are presented in Figs. 12 and 13. From these results, we can observe that most objects can be segmented from the fusion results, but they still have shortcomings, such as the objects are incomplete, which can be seen in the marked green and red boxes of Figs. 12 and 13, even the curves shown in Figs. 12(f) and 13(d) to 13(j), Bump shown in Fig. 13(i), and Bike shown in Figs. 13(d), 13(e), 13(f), and 13(g) are missing. By contrast, segmentation results on our fusion images are more accurate and much closer to the ground-truth maps due to that semantic information of the source images is used in our fusion strategy (as shown in Figs. 12(m) and 13(m)).

Additionally, we also provide their segmentation accuracies as shown in Table V. Here, the pixel differences of nine categories are measured using intersection-over-union (IoU) to assess their segmentation performance. Compared to fusion results produced by the other SOTA methods, the IBisNet network achieves the best accuracy of all categories on our fusion results, as well as the mean of IoU (mIoU). Such comparisons are the same as those in qualitative discussions and further demonstrate that our fusion results are more favorable for the segmentation task.

2) *Detection Task*: We also conducted object detection experiments using YOLOv5 [86] on the fusion results to validate their strong advantages in widespread applications. Figures 14 and 15 present two typical scenarios of visualized examples of object detection. From these examples, we can observe that although the compared fusion algorithms can improve image quality, their fusion results still have a lot of issues which had been analyzed in Sub-sect. IV-C and not beneficial for object detection, such as low accuracy (human in Fig. 14(f) and car in Fig. 15(d)) and object missing (bicycle in all the fusion results except our fusion result of Fig. 14 and car in Figs. 15(e), 15(h), and 15(i)). Although the bicycle in the “00638D” scenario (Fig. 14) is small in the dark area, YOLOv5 still accurately detect it from our fusion result,

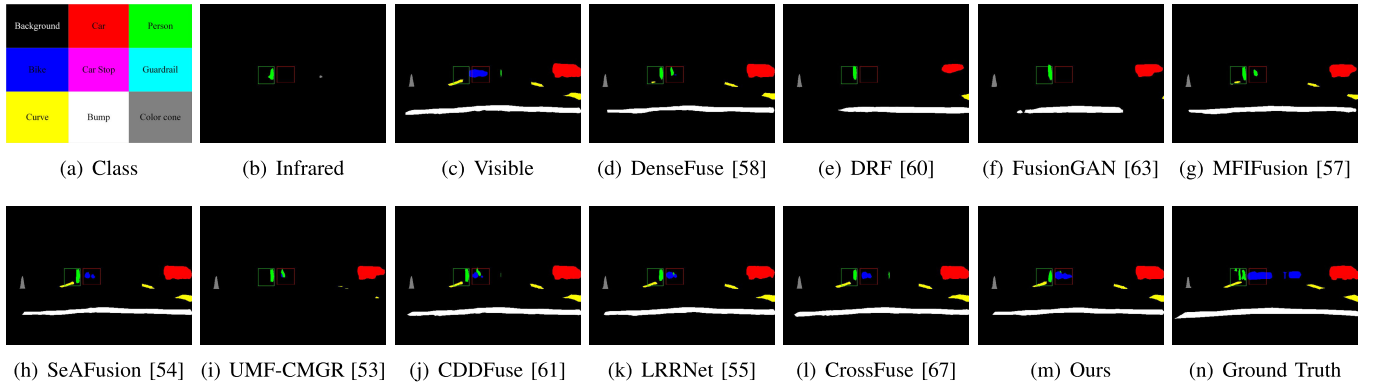


Fig. 13. Segmentation results with the IBisNet network on the fused images generated by applying the selected SOTA fusion methods to the “00504D” image from the *MSRS* dataset.

TABLE V

QUANTATIVE SEGMENTATION VALUES BY APPLYING THE IBISNET NETWORK TO VISIBLE, INFRARED AND FUSED IMAGES FROM THE *MSRS* DATASET. THE BEST ACCURACY IS MARKED WITH **RED** FONT WHILE THE SECOND ONE IS MARKED WITH **BLUE** FONT

	Background	Car	Person	Bike	Curve	Car Stop	Guardrail	Color Tone	Bump	mIoU
Visible	97.83	78.78	44.47	62.48	57.13	65.96	80.14	56.19	76.37	68.82
Infrared	96.01	60.46	60.60	34.22	46.07	33.54	65.57	26.88	42.67	51.78
DenseFuse [58]	98.44	82.72	63.81	60.37	60.12	65.38	77.21	55.86	78.66	71.40
DRF [60]	98.06	77.92	62.81	58.11	50.72	59.57	63.14	55.32	61.11	65.20
FusionGAN [63]	98.06	78.27	62.99	58.04	47.90	62.01	61.64	52.46	55.47	64.09
MFIFusion [57]	98.38	81.71	63.51	61.02	60.00	63.48	76.89	54.05	73.06	70.23
SeAFusion [54]	98.47	82.49	64.21	61.09	61.92	65.12	76.95	55.23	78.32	71.87
UMF-CMGR [53]	98.04	78.70	60.67	56.29	48.99	58.62	69.69	52.28	65.89	65.46
CDDFuse [61]	98.46	82.36	64.07	60.69	61.43	65.39	75.91	55.04	76.18	71.11
LRRNet [55]	98.23	80.53	58.46	59.19	53.57	64.45	79.53	55.71	76.04	69.97
CrossFuse [67]	98.33	80.59	59.97	61.63	59.49	55.50	78.21	55.49	77.10	69.59
Ours	98.47	83.21	64.40	61.96	62.06	65.47	79.95	55.93	79.05	72.28

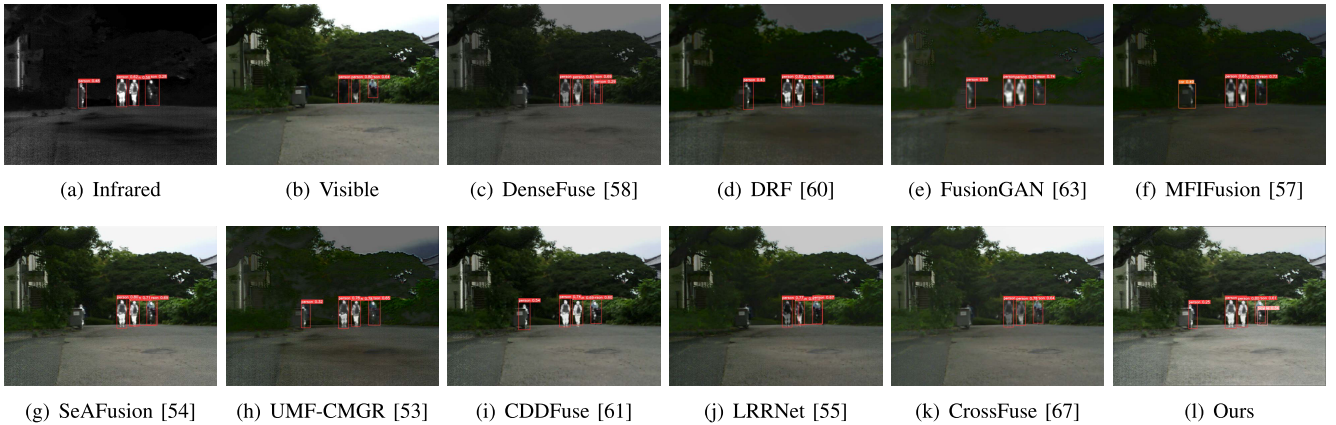


Fig. 14. Detection results with the YOLOv5 network on the fused images generated by applying the selected SOTA fusion methods to the “00637D” image from the *MSRS* dataset.

validating that our T²EA method performs well on preserving small-scale objects. And in the “00689N” night-time scene (Fig. 15), the high-contrast scenario with rich details and highlighted objects is helpful for YOLOv5 to improve detection accuracy. Meanwhile, the values of quantitative indicators including precision, recall, and mean average precision (mAP) are presented in Table VI to evaluate detection performance, where mAP@0.5, mAP@0.7, and mAP@0.9 represent the mAP values at IoU thresholds of 0.5, 0.7, and 0.9, respectively. From the comparisons, one can observe that the values

generated by YOLOv5 at each IoU threshold on our fusion results are the highest compared to those on other fusion methods. Such conclusion is the right response of that in the visual analysis and also illustrates that semantic segmentation is an important contributor in our T²EA network to improve the fusion performance.

F. Computational Complexity

To test their efficiency in terms of their runtime and the number of parameters, we conducted experiments on the

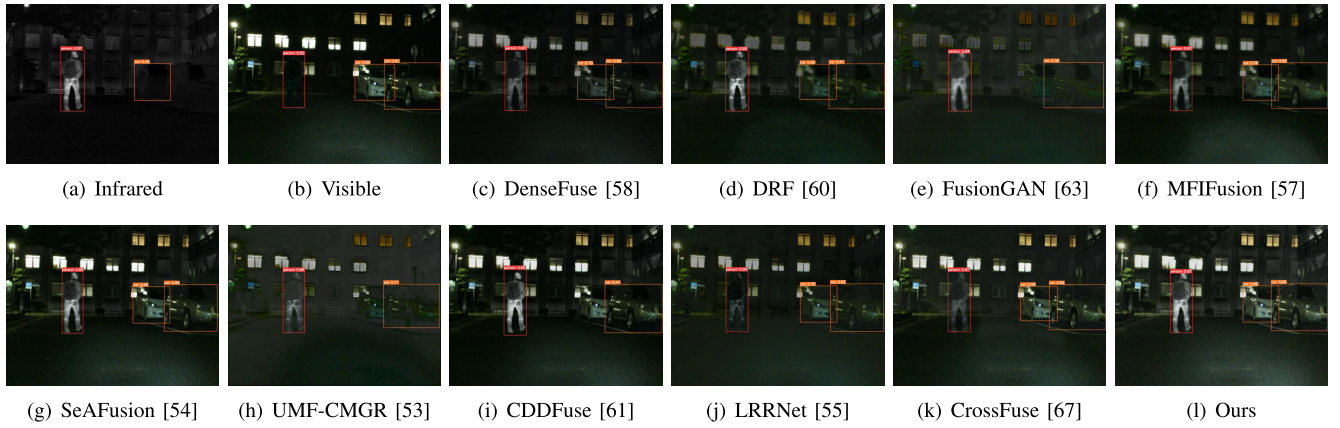


Fig. 15. Detection results with the YOLOv5 network on the fused images generated by applying the selected SOTA fusion methods to the “00689D” image from the *MSRS* dataset.

TABLE VI

QUANTITATIVE DETECTION VALUES BY APPLYING THE YOLOv5 NETWORK TO VISIBLE, INFRARED AND FUSED IMAGES FROM THE *MSRS* DATASET. THE BEST ACCURACY IS MARKED WITH **RED** FONT WHILE THE SECOND ONE IS MARKED WITH **BLUE** FONT. mAP@[0.5:0.95] DENOTES THE AVERAGE OF ALL mAP VALUES AT DIFFERENT IoU THRESHOLDS (FROM 0.5 TO 0.95 IN STEPS OF 0.05)

	mAP@0.5			mAP@0.7			mAP@0.9			mAP@[0.5:0.95]
	Precision	Recall	All	Precision	Recall	All	Precision	Recall	All	All
Visible	0.951	0.604	0.688	0.95	0.604	0.672	0.899	0.521	0.606	0.348
Infrared	0.929	0.904	0.922	0.861	0.943	0.668	0.917	0.763	0.863	0.611
DenseFuse [58]	0.932	0.868	0.926	0.923	0.85	0.916	0.845	0.764	0.837	0.597
DRF [60]	0.918	0.858	0.921	0.92	0.8	0.907	0.815	0.736	0.8	0.625
FusionGAN [63]	0.917	0.771	0.892	0.885	0.736	0.868	0.828	0.636	0.742	0.58
MFIFusion [57]	0.957	0.896	0.948	0.94	0.84	0.911	0.898	0.75	0.864	0.636
SeAFusion [54]	0.909	0.847	0.914	0.929	0.815	0.904	0.866	0.749	0.838	0.656
UMF-CMGR [53]	0.952	0.799	0.882	0.939	0.729	0.853	0.858	0.601	0.74	0.557
CDDFuse [61]	0.915	0.876	0.91	0.89	0.865	0.905	0.878	0.729	0.844	0.648
LRRNet [55]	0.954	0.856	0.918	0.943	0.851	0.91	0.843	0.735	0.845	0.653
CrossFuse [67]	0.921	0.823	0.899	0.913	0.816	0.885	0.865	0.659	0.811	0.607
Ours	0.961	0.925	0.953	0.961	0.896	0.952	0.906	0.769	0.867	0.669

TABLE VII

COMPARISON OF AVERAGE RUNNING TIME (UNIT: SECONDS)

Methods	Datasets			Parameters
	<i>TNO</i>	<i>LLVIP</i>	<i>MSRS</i>	
DenseFuse [58]	2.12	0.56	3.63	74,193
DRF [60]	18.47	26.96	70.33	48,139,285
FusionGAN [63]	2.98	2.66	7.89	1,326,404
MFIFusion [57]	4.76	0.91	3.12	2,275,558
SeAFusion [54]	1.03	0.55	2.66	166,883
UMF-CMGR [53]	6.12	1.91	6.99	629,253
CDDFuse [61]	2.09	1.62	4.92	1,188,272
LRRNet [55]	0.86	0.45	2.33	49,204
CrossFuse [67]	3.16	2.86	5.12	23,153,509
Ours	1.12	0.42	2.98	275,180

three datasets. In Table VII, we compare their time cost and the number of their used parameters. As one can see from the table, LRRNet achieves the best performance except on the *LLVIP* dataset with the smallest parameters. In contrast, our T²EA network is a little slower than LRRNet and SeAFusion except on the *LLVIP* dataset. However, by taking their remarkable fusion results for vision tasks into consideration,

our T²EA approach is still a competitive method for extensive applications.

V. CONCLUSION

It is a long-standing and prevailing study to integrate the highlighted objects in infrared images into the real-world scenarios of visible images for object detection. To achieve desired fusion results, this paper had reported a target-aware Taylor expansion approximation T²EA network for infrared and visible image fusion. The developed T²EA scheme includes three networks of Taylor expansion approximation (TEA), dual-branch feature fusion (DBFF), and semantic segmentation. Each of them has their own responsibility, such as TEA for image decomposition and feature maps composition, DBFF for decomposed maps fusion, and semantic segmentation for fusion feature refinement. Their architectures and loss functions had been detailly introduced in Sect. III. In the experiments, we first discussed the impact of the selection of Taylor expansion layers and fusion strategies on the fusion results. Then, we conducted several comparison experiments to validate the advantage of the proposed T²EA strategy, for instance, quantitative and qualitative comparisons to verify its excellent fusion performance, generalization experiments to validate its powerful capacity in image adaption, extensive applications to demonstrate its high visibility for vision tasks,

and computational complexity testing to illustrate its high efficiency. Our method also has a limitation that the TEA network and the segmentation network should be pre-trained when training the DBFF network due to the limitation of the computational capacity in our laboratory's computer device, which may insufficiently improve its fusion performance and may be overcome by employing a high-performance computer with one-stage training.

ACKNOWLEDGMENT

The authors would like to thank the editors and the anonymous reviewers for their valuable suggestions.

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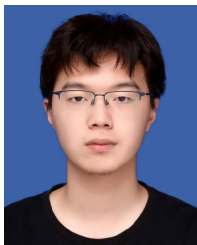
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